SL-I , W14

Linear SVM (non-separable case)

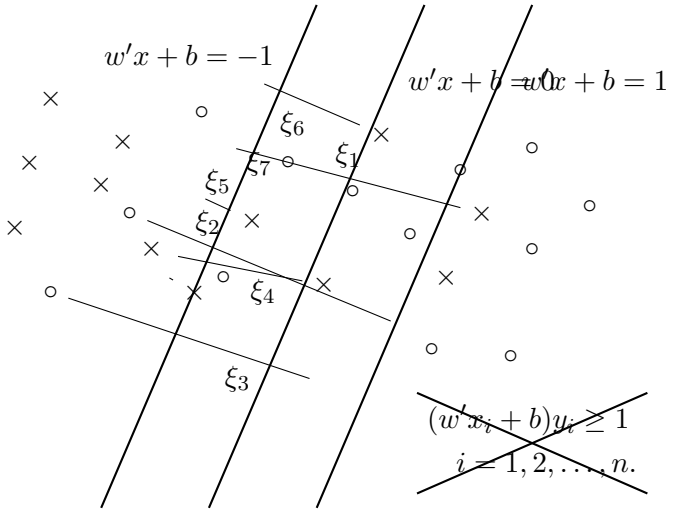


Figure 1: Linear SVM (non-separable case) with slack variables ξ_i.

In the non-separable case, we introduce some slack variables $\xi_1, \xi_2, \ldots, \xi_n \geq 0$ such that

$$(w'x_i + b)y_i \geq 1 - \xi_i$$

is satisfied for all $i = 1, 2, \ldots, n$

$\xi_i > 1 \iff$ The observation x_i is being misclassified

So, $\sum \xi_i$ serves as an upper bound for the no of misclassification.

Separable case,

$$\text{We minimize} \quad \frac{1}{2}\|w\|^2 \quad \text{s.t.} \quad (w'x_i + b)y_i \geq 1 \quad \forall i$$

Non-separable case,

PRIMAL

$$\begin{aligned} \text{We minimize} \quad & \frac{1}{2}\|w\|^2 + C\sum \xi_i \\ \text{s.t.} \quad & (w'x_i + b)y_i \geq 1 - \xi_i \quad \forall i = 1, 2, \ldots, n \\ & \xi_i \geq 0 \quad \forall i = 1, 2, \ldots, n \end{aligned}$$

$C > 0$ is a cost parameter. Larger values of C put more importance on the no of misclassification (hard thresholding) whereas the use of smaller values of C puts more importance on the margin maximization (soft thresholding).

Lagrangian

$$L\left(w, b, \xi_1, \ldots, \xi_n, \underline{\underline{\lambda_1, \ldots, \lambda_n, \mu_1, \ldots, \mu_n}}\right) = \frac{1}{2}\|w\|^2 + C\sum \xi_i + \sum \lambda_i \left[1 - \xi_i - (w'x_i + b)y_i\right] - \sum \mu_i \xi_i$$

Setting partial derivatives to zero:

$$\frac{\partial L}{\partial w} = 0 \implies \underset{\sim}{w} - \sum \lambda_i \underset{\sim i}{x}\, y_i = 0$$

$$\implies \underset{\sim}{w} = \sum \lambda_i \underset{\sim i}{x}\, y_i \tag{1}$$

$$\frac{\partial L}{\partial b} = 0 \implies$$

$$\sum \lambda_i y_i = 0 \tag{2}$$

$$\frac{\partial L}{\partial \xi_i} = 0 \implies C - \lambda_i - \mu_i = 0$$

NEW

$$\implies C = \lambda_i + \mu_i \quad (i = 1, 2, \ldots, n) \tag{3}$$

Derivation of the Dual Problem

Substituting back into the Lagrangian:

$$\begin{aligned}
&\frac{1}{2}\|w\|^2 + C\sum \xi_i + \sum \lambda_i \left\{1 - \xi_i - (w'x_i + b)y_i\right\} - \sum \mu_i \xi_i \\
&= \frac{1}{2}\left(\sum \lambda_i \underset{\sim i}{x}\, y_i\right)^T \left(\sum \lambda_i \underset{\sim i}{x}\, y_i\right) + \boxed{+C\sum \xi_i} \\
&\quad + \sum \lambda_i \boxed{-\sum \lambda_i \xi_i} \\
&\quad - \sum \lambda_i \left(\sum \lambda_i \underset{\sim i}{x}\, y_i\right)^T x_i y_i \\
&\quad - \sum \lambda_i b y_i \boxed{-\sum \mu_i \xi_i}
\end{aligned}$$

$$
\begin{aligned}
&= \frac{1}{2}\left(\sum \lambda_i x_i y_i\right)^T \left(\sum \lambda_i x_i y_i\right) + \sum \lambda_i \\
&\quad + \underbrace{\sum (C - \lambda_i - \mu_i)\xi_i}_{0} \\
&\quad - \left(\sum \lambda_i x_i y_i\right)^T \left(\sum \lambda_i x_i y_i\right) \\
&\quad - b \underbrace{\sum \lambda_i y_i}_{0} \\
&= \sum \lambda_i - \frac{1}{2}\left(\sum \lambda_i x_i y_i\right)^T \left(\sum \lambda_i x_i y_i\right) \\
&= \sum \lambda_i - \frac{1}{2} \sum_i \sum_j \lambda_i \lambda_j \langle x_i, x_j \rangle y_i y_j
\end{aligned}
$$

So, the dual problem is given by:

$$
\begin{aligned}
\max \quad & \sum \lambda_i - \frac{1}{2} \sum_i \sum_j \lambda_i \lambda_j \langle x_i, x_j \rangle y_i y_j \\
\text{s.t.} \quad & \sum \lambda_i y_i = 0 \\
& 0 \le \lambda_i \le C \quad \text{for all } i = 1, 2, \ldots, n.
\end{aligned}
$$

Solving for w^* and b^* using KKT Conditions

Let $\lambda_1^*, \lambda_2^*, \ldots, \lambda_n^*$ be the solution of the dual problem (obtained using a suitable convex opt algorithm)

$$
w^* = \sum \lambda_i^* \underset{\sim}{x}_i y_i
$$

b^* can be obtained by appropriately using the KKT conditions.

KKT condition (complementary slackness)

- $\odot$ $\lambda_i \left(1 - \xi_i - (w' x_i + b) y_i\right) = 0 \quad$ for all $i = 1, 2, \ldots, n$
- $\odot$ $\mu_i \xi_i = 0 \quad$ for all $i = 1, 2, \ldots, n$

Choose an i_0 such that

$$
0 < \lambda_{i_0} < C
$$

For such a choice we have

$$
\begin{aligned}
& \mu_{i_0} > 0 \quad (\lambda_{i_0} + \mu_{i_0} = C) \\
\implies & \xi_{i_0} = 0 \quad (\text{since } \mu_{i_0} \xi_{i_0} = 0)
\end{aligned}
$$

and hence

$$
\begin{aligned}
\lambda_{i_0} \left(1 - \underset{\overline{\overline{0}}}{\xi_{i_0}} - (w^{*\prime} x_i + b^*) y_i\right) &= 0 \\
\implies 1 - (w^{*\prime} x_i + b^*) y_i &= 0 \\
\implies b^* \text{ can be obtained from here.}
\end{aligned}
$$

In practice, we consider all such i's for which $\lambda_i \in (0, C)$ and compute the average of the corresponding b.

To classify a future obs x we compute

$$\begin{aligned} w^{*\prime}x + b^* &= \left(\sum \lambda_i^* x_i y_i\right)^T x + b^* \\ &= \sum \lambda_i^* \langle x_i, x\rangle y_i + b^* \end{aligned}$$

and classify to class-1 (resp class-2) if this quantity is pos (resp neg).

Nonlinear classification using SVM

Main idea - Project the observations to a higher dimensional space and construct a linear classifier in that space.

$$\begin{bmatrix} \underset{\sim}{x}_1 & y_1 \\ \vdots & \vdots \\ \underset{\sim}{x}_n & y_n \end{bmatrix} \longrightarrow \begin{matrix} \left(x_1^{(1)}, x_1^{(2)}\right) & y_1 \\ \left(x_2^{(1)}, x_2^{(2)}\right) & y_2 \\ \vdots & \vdots \\ \left(x_n^{(1)}, x_n^{(2)}\right) & y_n \end{matrix}$$

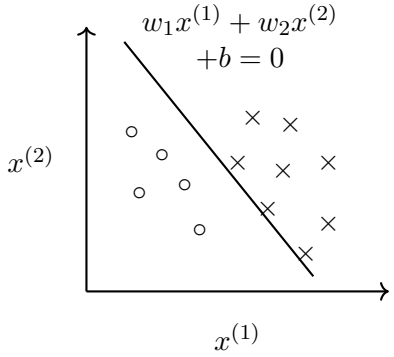


Figure 2: Linear separator $w_1x^{(1)} + w_2x^{(2)} + b = 0$ in 2D feature space.

$$\begin{matrix} \left(x^{(1)}, x^{(2)}\right) & \longrightarrow & \left(x^{(1)}, x^{(2)}, \left(x^{(1)}\right)^2, \left(x^{(2)}\right)^2, x^{(1)}x^{(2)}\right) \\ \mathbb{R}^2 & & \mathbb{R}^5 \\ \underset{\sim}{x} & \longmapsto & \underset{\sim}{x}^* \end{matrix}$$

$$\begin{bmatrix} x_1^* & y_1 \\ \vdots & \vdots \\ x_n^* & y_n \end{bmatrix}$$

A separating surface in the projected space is of the form

$$\begin{aligned} w_1x^{(1)} + w_2x^{(2)} + w_3\left(x^{(1)}\right)^2 + w_4\left(x^{(2)}\right)^2 \\ + w_5x^{(1)}x^{(2)} + b = 0 \end{aligned}$$

Linear:	$w_1x_1 + w_2x_2 = 0$
Quadratic:	$w_1x_1 + w_2x_2 + w_3x_1^2 + w_4x_2^2 + w_5x_1x_2 = 0$

$$\begin{array}{l|l} \underset{\sim}{X} = (x_1, x_2) \longrightarrow X^* = (x_1, x_2, x_1^2, x_2^2, x_1x_2) & \langle x_i, x_j \rangle \\ \underset{\sim}{Y} = (y_1, y_2) \longrightarrow Y^* = (y_1, y_2, y_1^2, y_2^2, y_1y_2) & \langle x_i, x \rangle \\ \langle X, Y \rangle \qquad\qquad \langle X^*, Y^* \rangle & \langle x_i^*, x_j^* \rangle \\ & \langle x_i^*, x^* \rangle \end{array}$$

Computing the inner product in the projected space:

$$\begin{aligned} \langle X^*, Y^* \rangle &= x_1y_1 + x_2y_2 + x_1^2y_1^2 + x_2^2y_2^2 \\ &\quad + x_1x_2y_1y_2 \end{aligned}$$

If we scale the coordinates with $\sqrt{2}$:

$$X = (x_1, x_2) \longrightarrow X^{**} = \left(\sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1x_2\right)$$
$$Y = (Y_1, Y_2) \longrightarrow Y^* = \left(\sqrt{2}Y_1, \sqrt{2}Y_2, Y_1^2, Y_2^2, \sqrt{2}Y_1Y_2\right)$$

$$\begin{aligned} \langle X^{**}, Y^{**} \rangle &= 2x_1Y_1 + 2x_2Y_2 + x_1^2Y_1^2 + x_2^2Y_2^2 \\ &\quad + 2x_1x_2Y_1Y_2 \\ &= (1 + x_1Y_1 + x_2Y_2)^2 - 1 \end{aligned}$$

$$\langle X^{**}, Y^{**} \rangle = \underbrace{(1 + \langle X, Y \rangle)^2 - 1}_{K(X,Y)}$$

$$\begin{array}{ccc} X & \longrightarrow & \phi_X \ (= X^*/X^{**}) \\ \mathbb{R}^d & & \text{Reproducing Kernel} \\ \text{Original space} & & \text{Hilbert Space (RKHS)} \\ Y & \longrightarrow & \phi_Y \end{array}$$

$$\langle \phi_X, \phi_Y \rangle_{\mathcal{H}} = K(X, Y) \longleftarrow \text{Aim}$$

What kind of kernels $K(x, y)$ can be used?

$$K(x, y) \qquad \begin{array}{ll} x \longrightarrow \phi_x & \langle \phi_x, \phi_y \rangle \\ y \longrightarrow \phi_y & = K(x, y) \end{array}$$

What are the reproducing kernels?
Ans:- The kernels which are positive semi-definite.

Positive Semi-Definite Kernels

Take any $m \in \mathbb{N}$ and $x_1, \ldots, x_m \in \mathbb{R}^d$.
Consider the matrix

$$\begin{bmatrix} K(x_1, x_1) & K(x_1, x_2) & \cdots & K(x_1, x_m) \\ K(x_2, x_1) & K(x_2, x_2) & \cdots & K(x_2, x_m) \\ \vdots & \vdots & \ddots & \vdots \\ K(x_m, x_1) & K(x_m, x_2) & \cdots & K(x_m, x_m) \end{bmatrix}$$

The matrix should be p.s.d: for any $a_1, a_2, \ldots, a_m \in \mathbb{R}$

$$\sum_i \sum_j a_i a_j K(x_i, x_j) \geq 0$$

If $K(x_i, x_j) = \langle \phi_{x_i}, \phi_{x_j} \rangle$ for some ϕ it is easy to show that K is p.s.d

Proof:

$$\begin{aligned}\sum_i \sum_j a_i a_j K(x_i, x_j) &= \sum_i \sum_j a_i a_j \langle \phi_{x_i}, \phi_{x_j} \rangle \\ &= \left\langle \sum_i a_i \phi_{x_i}, \sum_j a_j \phi_{x_j} \right\rangle \\ &= \left\| \sum_i a_i \phi_{x_i} \right\|^2 \geq 0.\end{aligned}$$

If $K(x, y) = \langle \phi_x, \phi_y \rangle$ for some ϕ, K is p.s.d.
The converse is also true.
If K is p.s.d, $\exists$ a transformation ϕ such that $K(x, y) = \langle \phi(x), \phi(y) \rangle$ for all $x, y \in \mathbb{R}^d$.

For nonlinear SVM, we use kernels which are:

i) symmetric $K(x, y) = K(y, x)$

ii) and positive semi-definite

Example 1 (Linear Kernel):

$$\begin{aligned}\underset{\sim}{X} &= (x_1, x_2) \qquad & \underline{K(X, Y)} &= \underline{x_1 Y_1 + x_2 Y_2} \\ \underset{\sim}{Y} &= (Y_1, Y_2) & &= \langle \underset{\phi_X}{X}, \underset{\phi_Y}{Y} \rangle\end{aligned}$$

Example 2 (Quadratic Kernel):

$$\begin{aligned}X &= (x_1, x_2) \\ Y &= (Y_1, Y_2) \\ K(X, Y) &= (1 + \langle X, Y \rangle)^2 \\ &= (1 + x_1 Y_1 + x_2 Y_2)^2 \\ &= 1 + 2x_1 Y_1 + 2x_2 Y_2 + x_1^2 Y_1^2 + x_2^2 Y_2^2 + 2x_1 x_2 Y_1 Y_2 \\ &= \left\langle \left(1, \sqrt{2}x_1, \sqrt{2}x_2, x_1^2, x_2^2, \sqrt{2}x_1 x_2\right), \left(1, \sqrt{2}Y_1, \sqrt{2}Y_2, Y_1^2, Y_2^2, \sqrt{2}Y_1 Y_2\right) \right\rangle \\ &= \langle \phi_X, \phi_Y \rangle \\ \implies & K \text{ is p.s.d}\end{aligned}$$

Some popular kernel functions for nonlinear SVM

i) **quadratic kernel**

$$K(\underset{\sim}{x}, \underset{\sim}{y}) = (1 + \langle \underset{\sim}{x}, \underset{\sim}{y} \rangle)^2$$

ii) **polynomial kernel**

$$K(\underset{\sim}{x}, \underset{\sim}{y}) = (1 + \langle x, y \rangle)^d$$

iii) **radial basis kernel**

$$K(x, y) = e^{-\gamma \| \underset{\sim}{x} - \underset{\sim}{y} \|^2} \quad \text{where } \gamma > 0$$

Linear SVM $\begin{cases} \longrightarrow \text{Separable case} \\ \longrightarrow \text{Non-separable case} \end{cases}$

Nonlinear SVM $\longrightarrow$ Project the obs to a higher dim space (RKHS) and perform lin classification there.

Support Vectors in Separable and Non-Separable Cases

$$\lambda_1^*, \ldots, \lambda_n^*$$
$$w^* = \sum \lambda_i^* \underset{\sim}{x}_i y_i$$

KKT

$$\lambda_i \left(1 - (w' x_i + b) y_i\right) = 0$$
$$\text{for all } i = 1, \ldots, n$$

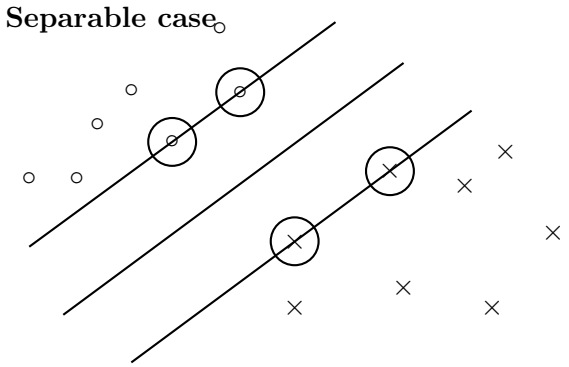


Figure 3: Support vectors (circled) in the separable case.

Non-separable case

Which are the support vectors?

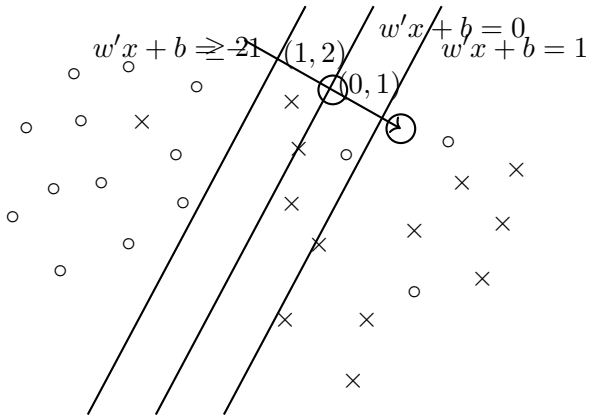


Figure 4: Analyzing support vectors in the non-separable case.

KKT

$$\lambda_i \left\{1 - \xi_i - (w'x_i + b)y_i\right\} = 0 \quad \text{for all } i = 1, \ldots, n$$
$$\mu_i \xi_i = 0 \quad \text{for all } i = 1, \ldots, n.$$

$$C = \mu_i + \lambda_i \quad \text{for all } i = 1, 2, \ldots, n$$

$$\lambda_i = 0 \implies \mu_i = C \implies \xi_i = 0$$

Observations with $\xi_i > 0$ are support vectors.

Whenever $\xi_i > 0$, we shall also have $\lambda_i > 0$

$\lambda_i = 0$	$\xi_i = 0$	
$\lambda_i > 0$	$\xi_i = 0$	
$\lambda_i = 0$	$\xi_i > 0$	$\times$

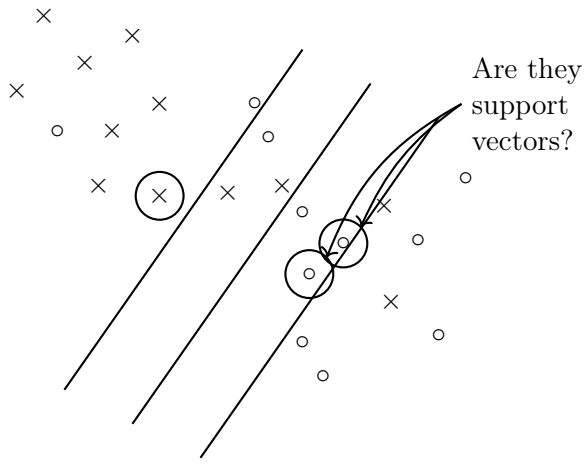


Figure 5: Support vectors lying on the margin ($\xi_i = 0, \lambda_i > 0$) and violating the margin ($\xi_i > 0, \lambda_i = C$).

Nonlinear SVM: Original Space vs. Projected Space

x_1	$\cdots$	x_n
y_1	$\cdots$	y_n

Original space ($\mathbb{R}^d$) $\quad (x \longrightarrow \phi_x)$

ϕ_{x_1}	$\cdots$	ϕ_{x_n}
y_1	$\cdots$	y_n

Projected space
($\mathcal{H}$)
$\downarrow$
Hilbert Space

Primal Formulation Comparison

Original space	**Projected space**
PRIMAL	**PRIMAL** ??
Min $\frac{1}{2}\|w\|^2 + C\sum \xi_i$	Min $\frac{1}{2}\langle w, w\rangle + C\sum \xi_i$
s.t. $(w'x_i + b)y_i \geq 1 - \xi_i$	s.t. $\langle w, \phi_{x_i}\rangle y_i \geq 1 - \xi_i$
for all $i = 1, 2, \ldots, n$	for all $i = 1, 2, \ldots, n$
$(w \in \mathbb{R}^d)$	$(w \in \mathcal{H})$

Dual Formulation Comparison

Original Space	**Projected space**
DUAL	**DUAL**
Max	Max
$\sum \lambda_i - \sum_{i,j} \lambda_i\lambda_j\langle x_i, x_j\rangle y_i y_j$	$\sum \lambda_i - \sum_{i,j} \lambda_i\lambda_j K(\underset{\sim}{x}_i, \underset{\sim}{x}_j) y_i y_j$
s.t. $\sum \lambda_i y_i = 0$	s.t. $\sum \lambda_i y_i = 0$
$\lambda_i \geq 0 \quad \forall i$	$\lambda_i \geq 0 \quad \forall i$
$\lambda_i \leq C \quad \forall i$	$\lambda_i \leq C \quad \forall i$

Computing w^* and b^* in Projected Space

Suppose that $(\lambda_1^*, \ldots, \lambda_n^*)$ is the solution of the dual:

Original space	**Projected space**
?? $w^* = \sum_{i=1}^n \lambda_i^* \underset{\sim}{x}_i y_i$	$w^* = \sum_{i=1}^n \lambda_i \phi_{x_i} y_i$
We look for i_0 such that $0 < \lambda_{i_0} < C$.	3=⟶ Is it possible to adapt this method to get b?
For such an i_0, we have $(w'x_{i_0} + b)y_{i_0} = 1$.	
b^* is obtained from here.	

Consider an i_0 such that $0 < \lambda_{i_0}^* < C$.
For such an i_0, we have

$$\begin{aligned}
&\left[\langle w^*, \phi_{x_{i_0}}\rangle_{\mathcal{H}} + b\right] y_{i_0} = 1 \\
\Longrightarrow &\left[\left\langle \sum_{i=1}^n \lambda_i^* \phi_{x_i} y_i, \phi_{x_{i_0}} \right\rangle + b\right] y_{i_0} = 1 \\
\Longrightarrow &\left[\sum_{i=1}^n \lambda_i^* \langle \phi_{x_i}, \phi_{x_{i_0}}\rangle y_i + b\right] y_{i_0} = 1 \\
\Longrightarrow &\left[\sum_{i=1}^n \lambda_i^* K(x_i, x_{i_0}) y_i + b\right] y_{i_0} = 1
\end{aligned}$$

b^* can be computed from this equation.
As before, we can consider several such i_0's, compute b^* for each them and use the average value for all practical purposes.

Classification of a future observation x

Original space	**Projected space**
If $w^{*\prime}x + b^* > 0$:	Here we need to compute
Classify it to Class-1 (positive class)	$\langle w^*, \phi_x \rangle + b^*$
If $w'x + b < 0$:	$= \left\langle \sum_{i=1}^{n} \lambda_i^* \phi_{x_i} y_i, \phi_x \right\rangle + b^*$
Classify x to Class-2 (negative class)	$= \sum_{i=1}^{n} \lambda_i^* \langle \phi_{x_i}, \phi_x \rangle y_i + b^*$
	$= \sum_{i=1}^{n} \lambda_i^* K(x_i, x) y_i + b^*$

We assign x to class-1 (+ve class) if this quantity is positive. Otherwise, we assign it to class-2 (-ve class).

Even if we do not know ϕ explicitly, as long as the positive semi definite kernel K is known to us, we can construct a non-linear classifier using SVM.

Additional Examples of Kernel Feature Maps

Example A:

$$\begin{aligned}
\mathbb{R}^2: \quad X &= (x_1, x_2), \qquad Y = (Y_1, Y_2) \\
K(X,Y) &= x_1^2 Y_1^2 + x_2 Y_2 \\
&= \left\langle (x_1^2, x_2), (Y_1^2, Y_2) \right\rangle \\
&= \langle \phi_X, \phi_Y \rangle \\
(x_1, x_2) &\longrightarrow (x_1^2, x_2) \\
X &\qquad\quad \phi_X
\end{aligned}$$

Example B:

$$\begin{aligned}
K(X,Y) &= a_0 + a_1 x_1 Y_1 + a_2 x_2 Y_2, \quad \text{where } a_0, a_1, a_2 > 0 \\
&= \left\langle \underbrace{(\sqrt{a_0}, \sqrt{a_1} x_1, \sqrt{a_2} x_2)}_{\phi_X}, \underbrace{(\sqrt{a_0}, \sqrt{a_1} Y_1, \sqrt{a_2} Y_2)}_{\phi_Y} \right\rangle \\
(x_1, x_2) &\longrightarrow \left(\sqrt{a_0}, \sqrt{a_1} x_1, \sqrt{a_2} x_2\right) \\
X &\qquad\quad \phi_X
\end{aligned}$$

Example C:

$$\begin{aligned}
K(X,Y) &= (1 + \langle X, Y \rangle)^2 \\
&= (1 + x_1 Y_1 + x_2 Y_2)^2 \\
&= 1 + x_1^2 Y_1^2 + x_2^2 Y_2^2 + 2x_1 Y_1 + 2x_2 Y_2 + 2x_1 Y_1 x_2 Y_2 \\
\phi_X &= \left(1, x_1^2, x_2^2, \sqrt{2} x_1, \sqrt{2} x_2, \sqrt{2} x_1 x_2\right) \\
\phi_Y &= \left(1, Y_1^2, Y_2^2, \sqrt{2} Y_1, \sqrt{2} Y_2, \sqrt{2} Y_1 Y_2\right)
\end{aligned}$$

Logistic Regression

Two-class

$$\begin{aligned}
\log \frac{p_1(x)}{p_0(x)} &= \alpha' x + \beta \\
p_1(x) &= e^{\alpha' x+\beta} p_0(x) \\
p_1(x) &= \frac{e^{\alpha' x+\beta}}{1+e^{\alpha' x+\beta}} \\
p_0(x) &= \frac{1}{1+e^{\alpha' x+\beta}}
\end{aligned}$$

With y_i = Class label:

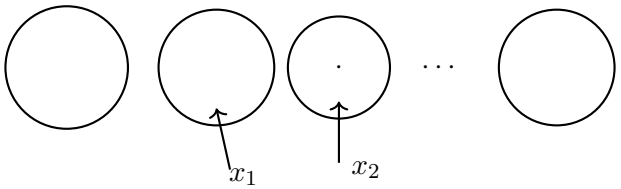
x_1	y_1
x_2	y_2
$\vdots$	$\vdots$
x_i	y_i
$\vdots$	$\vdots$
x_n	y_n

$$L = \prod_{i=1}^{n} \left(\frac{e^{\alpha' x_i+\beta}}{1+e^{\alpha' x_i+\beta}} \right)^{y_i} \left(\frac{1}{1+e^{\alpha' x_i+\beta}} \right)^{1-y_i}$$

Multi-class logistic regression

$$\left.\begin{aligned}
\log \frac{p_1(x)}{p_0(x)} &= \alpha_1' x + \beta_1 \\
\log \frac{p_2(x)}{p_0(x)} &= \alpha_2' x + \beta_2 \\
&\vdots \\
\log \frac{p_k(x)}{p_0(x)} &= \alpha_k' x + \beta_k
\end{aligned}\right\} \text{model}$$

$$\begin{aligned}
p_1(x) &= \frac{e^{\alpha_1' x+\beta_1}}{1+\sum_{i=1}^{k} e^{\alpha_i' x+\beta_i}} \\
p_2(x) &= \frac{e^{\alpha_2' x+\beta_2}}{1+\sum_{i=1}^{k} e^{\alpha_i' x+\beta_i}} \\
&\vdots \\
p_k(x) &= \frac{e^{\alpha_k' x+\beta_k}}{1+\sum_{i=1}^{k} e^{\alpha_i' x+\beta_i}} \\
p_0(x) &= \frac{1}{1+\sum_{i=1}^{k} e^{\alpha_i' x+\beta_i}}
\end{aligned}$$



$$\left\{ \; p_2(x_1),\; p_3(x_2) \; \right\}$$

Figure 6: Class probabilities assigned to observations x_1, x_2 across multiple classes.

SVM for multi-class problem

$\binom{k}{2}$

One vs One	For any given x
1 vs 2	1
1 vs 3	3
1 vs 4	4
2 vs 3	3
2 vs 4	2
3 vs 4	3

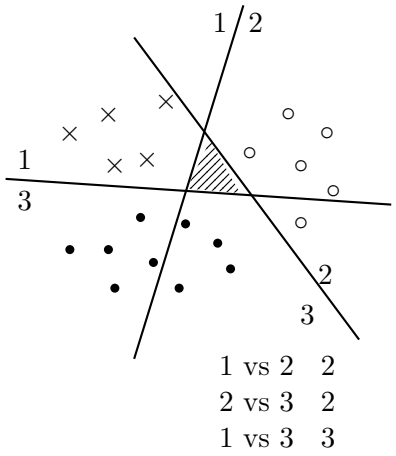
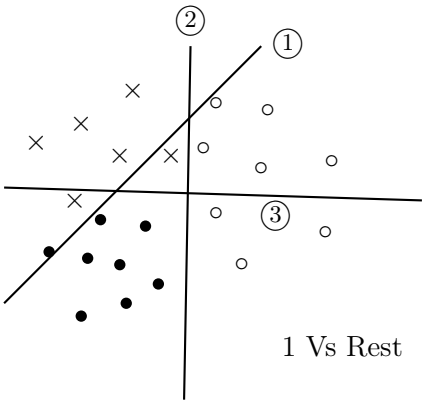


Figure 7: Pairwise (One vs One) hyperplanes and voting in 3-class SVM.

One vs One

Pairwise classification + Majority voting
Ties can be solved arbitrarily.
Otherwise, one can use the pairwise coupling method.
(Hastie, Tibshirani, 1998)

One vs Rest (1 Vs Rest)



1 Vs Rest

1×	2	3
1×	3×	2
3×	1	2

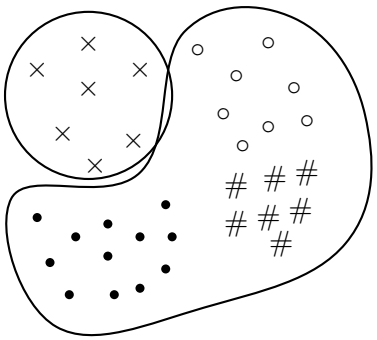


Figure 8: One vs Rest classification (left: linear boundaries and ambiguity regions; right: non-linear class vs. rest partitioning).